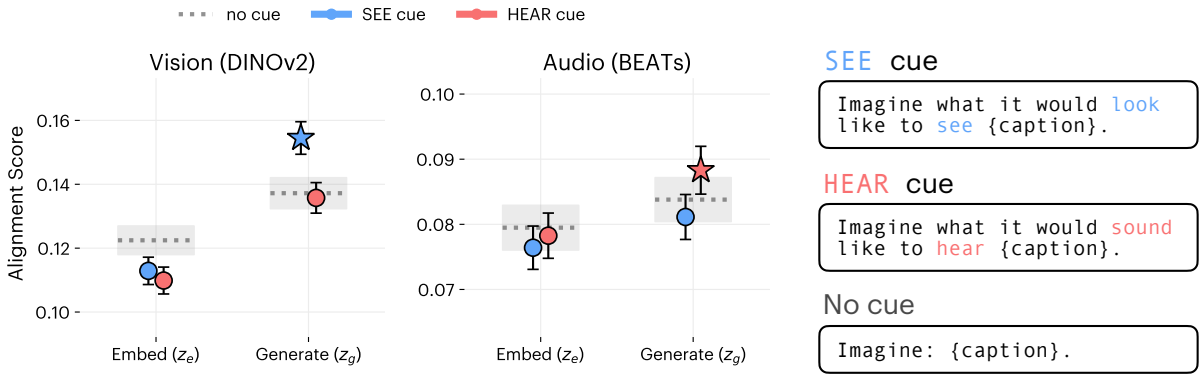


(a) Sensory cued alignment using single-pass embedding (z_e) and generative (z_g) representations.

(b) Prompt templates used in generative representations.

Figure 3: Sensory cues induce a generative text-only LLM representation that has higher alignment with the corresponding encoder. The star denotes matching cue-modality in generative representation.

2.4 Datasets

We evaluate on two image–caption and two audio–caption datasets. **WiT** [Srinivasan et al., 2021]: 1024 image–caption pairs from Wikipedia (as in Huh et al. [2024]). **DCI** [Urbanek et al., 2024]: 1024 summarized captions of densely captioned images. **AudioCaps2.0** [Kim et al., 2019]: 975 audio–caption pairs from AudioSet. **Clotho v2** [Drossos et al., 2020]: 975 audio–caption pairs from the evaluation split, one caption per clip.

3 Results

3.1 Generative Representations Yield Higher Alignment

We first compare alignment based on single-pass embeddings with alignment from generative representations, where the LLM continues each caption for 128 tokens under the no-cue template (Figure 3b). As shown in Figure 2, simply allowing the model to elaborate on the caption already produces embeddings that align more closely with sensory encoders.

This suggests that prior work such as Huh et al. [2024], which evaluated alignment only from single-pass embeddings, underestimate how much cross-modal similarity is present in LLMs. Generation creates a representation that yields higher alignment—even without explicit sensory cues. It offers a way to achieve such alignment at inference time, without requiring additional training.

3.2 Sensory Cues Steer Generative Representations

We find that explicit sensory cues in the prompt can controllably steer generative representations (z_g) to achieve higher alignment scores than under naive generation (i.e., when prompted without a sensory cue). We evaluate Qwen3-32B on paired image–text (WiT) and audio–text (AudioCaps) datasets by prepending each ground-truth caption with either a no cue baseline, a SEE cue, or a HEAR cue (Figure 3b). Each prompt generates a 128-token continuation, snippets of which are shown in Figure 4 (image-caption pairs from WiT).

Figure 3a shows that the SEE cue increases alignment with the vision encoder (DINOv2) and decreases alignment with the audio encoder (BEATs). Conversely, the HEAR cue increases alignment with BEATs while reducing alignment with DINOv2. These results indicate that a single sensory cue in the prompt can steer the internal representations of the LLM to better match the geometry of the modality the cue invokes. In contrast, we find that sensory prompting cannot steer single-pass embedding representations (z_e) in the same way. Inserting sensory cues into the prompt decreases the alignment from the no cue prompt. Thus, the higher alignment achieved through sensory prompting is a result of the representation formed during generation. We validate this result on additional sensory encoders, language models, and datasets in Appendix C.

Image / Caption		SEE cue generation	HEAR cue generation
	Awaiting a pitch; batter; catcher, and umpire in baseball	...I need to visualize a baseball field. The batter stands at the home plate, right? They're holding the bat, probably in a ready position, feet shoulder-width apart. Their body language should show they're focused, maybe squinting at the catcher or the umpire...	...There's the general ambient noise of the stadium: crowd murmurs, maybe some distant cheers or chatter. The crack of a bat from another field if it's a doubleheader or a minor league game nearby. The sound of players moving, like cleats scraping on the dirt, gloves being adjusted...
	Detail of the stained glass windows at the Chapel of St Saviour	...Let me start by recalling what I know about stained glass in chapels. Usually, they have intricate designs with biblical scenes, saints, maybe some geometric patterns. The colors are vibrant because of the glass, and when light passes through, it creates a colorful effect...	...Hmm, stained glass windows are usually visual, so translating that into sound is an interesting challenge ... glass, lead, maybe some metal reinforcements. Each material has its own acoustic properties. Glass can make a ringing sound when struck, like a bell...
	Groundnut kernel	...I should think about the color. When you open a peanut shell, the kernel is usually a pale yellow or light brown, maybe with a slight reddish tinge. But I'm not entirely sure if there are different varieties with different colors...	...Let me start by breaking down the process. When you crack a peanut shell, there's a distinct sound—crunchy, maybe a bit sharp. But the kernel itself, once the shell is off, is softer. If you bite into it, it's more of a soft crunch, maybe with some give...

Figure 4: Snippets of text generated from the caption, under sensory cues. *We highlight, by hand, words that may be associated with the sensory modality.* Full example found in Appendix G.1.

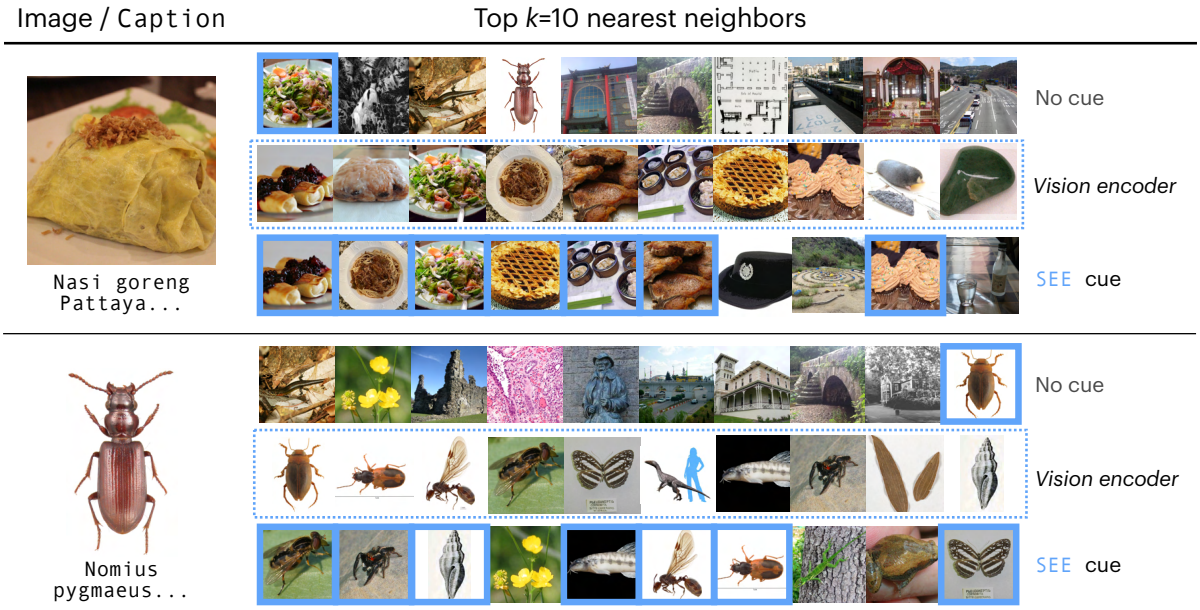


Figure 5: Selected examples where visual prompting yields the largest increase in shared top-k = 10 neighbors with the vision encoder (vs. no cue). Blue outlines mark inputs also among the vision encoder's nearest neighbors. Additional generations and examples appear in Appendix B.

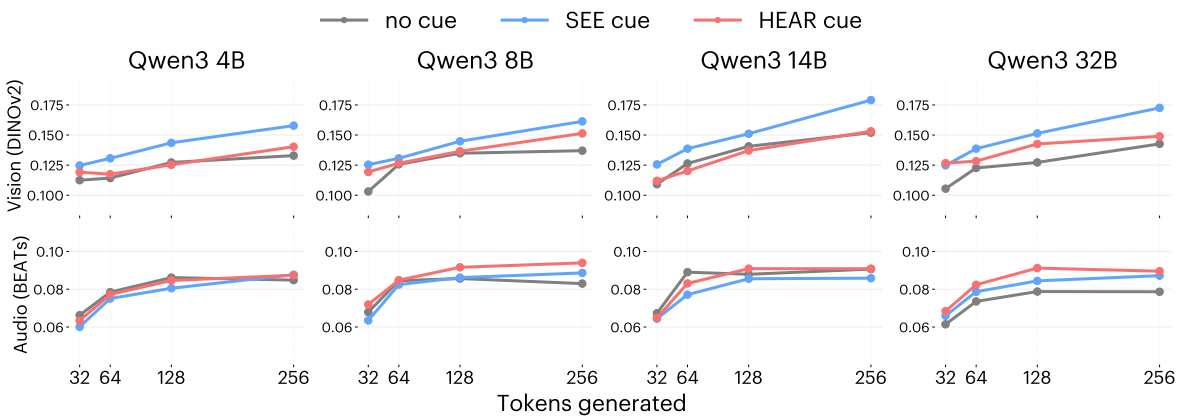
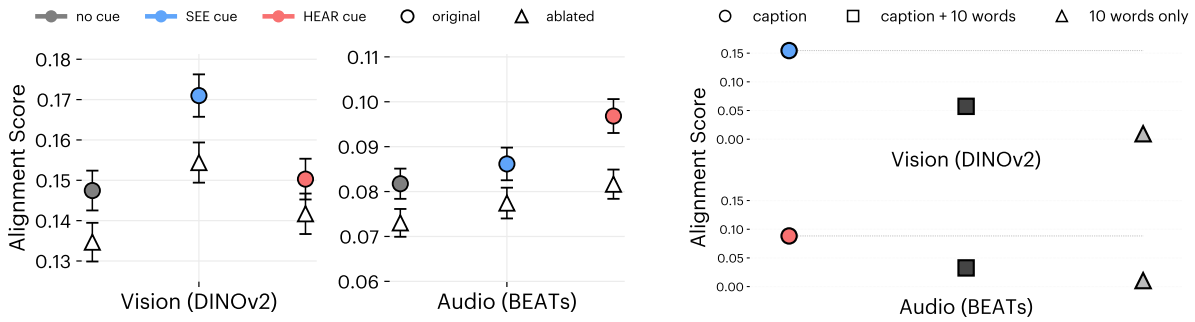


Figure 6: Alignment to sensory encoders increases with generation length.



(a) Sensory language ablation decreases alignment.

(b) Visual word hallucination control.

Sensory ablation

```
Please rewrite the following text by removing all sensory-specific words or descriptions
(e.g., related to sound, sight, smell, touch, taste) and replacing them with neutral, non-
sensory words; preserve the event or action while removing explicit sensory grounding: {text}.
```

(c) Prompt template

Figure 7: Correct sensory language is necessary for increase in alignment. Error bars not visible in (b) due to scale.

Mutual-kNN provides an interpretable illustration of when sensory prompting helps shift the LLM's representation toward the intended modality: Figure 5 shows two examples from WiT where the SEE cue yields the largest increase in shared top-k neighbors with the vision encoder. We consider the caption "Nasi goreng Pattaya." Under the no cue condition, the generation describes general information (e.g., *"Nasi goreng Pattaya is a local delicacy from Pattaya, Thailand, but it's also popular in neighboring countries"*), and the nearest neighbors include "Yam thale (Thai dish)", "Lankascincus gansi (a skink species in Sri Lanka)", and the "Korean–Chinese Cultural Center in Incheon, South Korea". These neighbors plausibly arise from the model's emphasis on geographic and cultural descriptors in the no cue text, which happen to be less visually related. By contrast, the SEE cue shifts the continuation toward concrete food description (*"the main components: fried rice, omelette, and the sauce… toppings like shrimp, chicken, or vegetables"*). Its nearest neighbors under this condition are themselves food items—such as "Blinchiki filled with cheese and topped with blackberries" and "Spaghetti topped with pulled pork in a marinara sauce..."—showing that by emphasizing visual descriptors of the food, the LLM produces a representation that aligns more closely with the vision encoder's representation of the corresponding image. Additional qualitative examples (including those that decrease the overlap in nearest neighbors) can be found in Appendix B.

3.3 Editing Sensory-Cued Generative Representations

Having shown that sensory cues condition generation such that its representation is more reflective of a given modality, we now explore which aspects of the generations are responsible for this effect. Generative hidden states are defined over the model's own outputs, and thus, editing the generation amounts to editing the representation itself. This allows us to test how alignment depends on specific language choices. We find that a *lack of sensory words* lowers alignment, but not just any sensory words increase it—what matters is using *scene-appropriate sensory details.*

Sensory-word ablation. To determine whether alignment depends on the explicit use of modality-specific language, we perform a sensory-word ablation on 256-token generations from Qwen3-32B using the prompts in Figure 7c. We choose longer generations to ensure that the original outputs contain sufficient sensory references for a meaningful intervention. Importantly, the ablation preserves the semantic content of each generation while replacing modality-specific language with neutral phrasing (see Appendix G.3 for examples). Following ablation, alignment to both vision and audio encoders drops significantly (Figure 7a), thus, sensory language is necessary for the observed alignment.